SoK: Reusing Defense Evidence in LLM and Agent Side Channels

Abstract—**Deploying an LLM or agent defense requires deciding which protection, function and cost evidence survives a change of context. This SoK organizes that decision around three boundaries: the receiving view, the affected execution state, and the attacker and workload. Source-linked evidence cards record supported observations, incompatible measurement objects and unresolved reuse questions; analyst benefit and coding reliability are not established. An attack-evaluated map of 19 mitigation families and a purposive six-source defense-first challenge set supply case material, not estimates of reporting prevalence. A frozen retrospective source map also admits formal and design evidence without requiring an empirical attack score. Selected cache and hardware examples distinguish configuration associations from paired measurements and conditional guarantees from empirical results. Coverage, source identity and eligibility remain incomplete. A finite synthetic replay study on three pinned local model artifacts illustrates why a clean-trained attack, an adapted attack and a calibration comparison require separate interpretation; its default identical-parameter controls provide no transfer evidence. The calibration comparisons do not establish a robust transfer effect. Fresh retail-sandbox interactions quantify a deliberately state-changing write condition in a nonrandom, mutable-service-alias cohort. These are bounded applications of established evaluation principles, not a validation of the reporting procedure or a general estimate of deployment harm.**

1. Introduction

Encryption conceals content while serving caches, packet streams and tool executions expose other observations [1], [2], [3]. A deployment can keep the name of a defense while changing what an observer sees, which state its cover affects, or how the attacker is trained. Choosing a mechanism is therefore only the start of a deployment decision. The next question is: *which protection, function and cost evidence can be retained across the proposed change?*

One defense name can refer to different evidence. CachePrune's selective-sharing results can inform a configuration-specific feasibility assessment [4, 5.1–5.6]. Early Bird's prefix policy and semantic-cache prototype require separate cost decisions [1, V-A–B]. These distinctions follow from the observer, primitive, operating point and measurement object. A shared mechanism or dataset name does not establish paired observations. The useful outcome is a scoped candidate and a measurement decision.

The systematization follows three boundaries.

- **Receiving view.** A protected representation can be followed by another emitter or observable. Retain the exact receiver and permitted fields, not a blanket privacy label.
- **Affected state.** Discarding a reply and preventing an action's effects are different operations. Retain a content-preservation witness where it applies; separately establish task outcomes for added execution.
- **Attacker and workload.** A clean-trained classifier, a defense-aware attacker and a source-calibrated defense establish different objects. Retain their fitting conditions and holdout units with their measured function and cost.

These boundaries organize the paper's contribution: a source-bound account of *defense-evidence reuse*. The evidence card records the protected target, receiver, intervention, attacker, function and cost together with their source locations. Record-level joins establish which measurements describe the same configuration; the transfer audit establishes what was held fixed or refitted. Their output is a proposed, inspectable assessment of what to retain, separate or remeasure (Table 1). No analyst comparison, independent coding study or deployment trial establishes that this procedure improves decisions, agreement or efficiency.

Source selection limits the empirical claims. The original auxiliary-channel search links 19 privacy/attribution mitigation families to evaluated records, including both attack-oriented and defense-oriented studies. A purposive defense-first challenge set adds four named seeds and two selected sources representing cache isolation, protected interfaces, communication shaping and agent privacy. Focused hardware readings further expose evidence outside that core frame. These sources inform the conceptual classification and practical decisions; they do not repair selection bias or support field-wide reporting rates. We keep their receiving views and measurement scopes separate; counts and source-removal sensitivities remain reconstructable from the cards.

Worked evidence makes the decisions testable. The synthesis builds on website-fingerprinting and adaptive-evaluation practice [5], [6], [7], [8]. Its LLM/agent-specific contribution connects prefix and semantic caches, authorized message content and representation choice, and discarded responses and executed effects. Two new studies make the latter decisions concrete. Fixed-weight generation followed by timed TLS loopback supplies an exploratory synthetic example of attacker adaptation, while its calibration audit distinguishes identical-parameter controls from conditional

TABLE 1. THREE PROPOSED EVIDENCE-REUSE QUESTIONS. THESE REPORTING DISTINCTIONS ARE ILLUSTRATED BY SOURCES AND WORKED CASES; ANALYST BENEFIT IS NOT EVALUATED.

Boundary	Evidence-reuse finding	Concrete decision
Receiving view	Upstream framing, authorized message content and internal scores protect different observations.	Capture the final receiver; retain only the fields covered by the result.
Affected state	Preserved replies and discarded cover outputs do not establish preserved execution; fresh sandbox tasks expose the consequences of admitted writes.	Retain scoped preservation; measure task completion, state changes and quota use under the executed intervention.
Attacker/workload	Adaptation and source-calibration transfer answer different questions; the E4 source-calibration comparisons are degenerate or inconclusive.	Fit the intended attacker on defended data; compare imported calibration with target-local fitting and retain the paired uncertainty.

source comparisons. Fresh three-arm retail-sandbox interactions measure task utility when an extra read or same-item write is hidden from the planner. The cases instantiate reporting distinctions rather than test their usefulness to analysts. Existing external applications comprise 15 measurement records for 14 defense–study pairs across three published WF studies; their outcomes were visible when coded. The synthesis presents selected source-to-assessment examples, while the supplement retains the full ledger. Experiment results and collection provenance remain in Appendix A and the supplement.

2. Sources and Evidence Cards

2.1. Selection frames and focused source follow-up

The original search, closed on October 21, 2026, maps 82 study families: 58 core and 24 adjacent. Core inclusion requires LLM/agent execution (C1), a hidden property (C2), an execution-dependent auxiliary observable (C3), specified observer permissions (C4) and an empirical inference evaluation (C5). Adjacent families remain contextual entries rather than cases satisfying all five conditions. Ordinary response semantics alone do not satisfy the auxiliary-observable condition. The resulting privacy/attribution mitigation sample is therefore selected through attack evaluation, not a census of defense literature. It includes both attack-oriented and defense-oriented studies.

A defense-first exercise on October 27 examines a purposive challenge set: four named seeds (CacheSolidarity, PlanTwin, SICC and DP-MemView) and two selected additions (OCELOT and AirGapAgent) from nine logged searches. Prior reconnaissance and the four-seed/two-addition cap are recorded, but the retained protocol has no independently verifiable preregistration timestamp. The exercise tests source-specific scope interpretations; it does not correct the original selection bias or supply a representative defense sample. The cards distinguish auxiliary timing/form/score observations from semantic transcript and information-flow protection. Source versions, search returns, admission decisions and unassessed candidates are retained in the supplement.

The original search uses seeds, venue-targeted screening, citation tracing, Semantic Scholar and native arXiv queries. The frozen map and query dispositions remain available for reconstruction. Neither it nor the challenge exercise systematically covers architecture/hardware venues such as ISCA, MICRO, HPCA, ASPLOS and CHES/TCHES. Formal guarantees, component costs and empirical recovery scores can answer different questions; requiring a scored auxiliary-channel attack also excludes some useful defense evidence. Privacy and attribution are the primary protected interests; evasion and authorized diagnostics remain boundary comparisons rather than pooled privacy defenses.

The retrospective R47 follow-up broadens eligibility beyond C5. Direct defenses (D) explicitly propose, implement, formally analyze or evaluate confidentiality or unlinkability against an auxiliary observation in LLM inference, serving or agent execution; neither an empirical attack-success result nor a particular R1–R9 binding is required. Relevant generic CPU/GPU, trusted execution environment (TEE), oblivious RAM (ORAM), constant-time and traffic mechanisms without explicit LLM/agent analysis remain mechanism background (M). Semantic-content privacy, general model safety and secondary discovery material without that auxiliary-observation claim remain adjacent (A); unresolved cases stay pending. Protected interests are recorded separately: user/task privacy, model/service secrets, attribution and audit/detection integrity do not share an effectiveness denominator.

Search and acquisition are frozen at the evidence already recorded for this revision. The protocol's discovery interval is January 1, 2018 through October 30, 2026, 13:15:46 UTC, with earlier foundations retained as background; later readings do not establish public availability at that cutoff. Partial venue inventories, capped index returns and selected-section readings leave coverage, version/family identity and final eligibility incomplete. Earlier results and reviews were visible during this nonblind, model-assisted work. It was neither preregistered nor independently human-coded. Selected sources support bounded comparisons, not a complete defense corpus or field-wide rates. The frozen C1–C5 denominators and historical codes remain unchanged.

2.2. Source-bound coding and scope

The retained R1–R9 codes and comparison decisions are the recorded outputs of the disclosed source-coding process; they do not imply independent human coding. R47 source interpretations and unresolved eligibility decisions remain

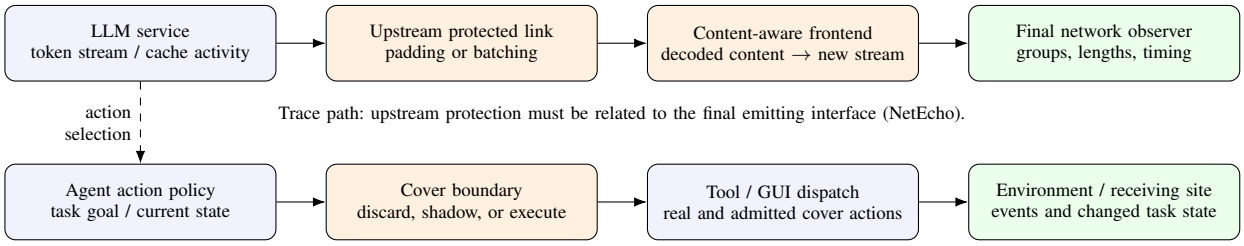


Figure 1. Receiving and execution boundaries. A new content-aware emitter can create another trace; admitted cover actions can change state. Evidence must accompany the object actually observed or executed.

subject to author review before finalization. Each card binds a source version and section/table/figure locator to its target, receiver, intervention, attacker/test condition, functional outcome and cost. Recorded revisions preserve the reason for a scope or applicability change. B denotes an explicit binding, P partial evidence, U unreported in the inspected scope, ? unverified, and – inapplicable to the requested outcome. Deterministic checks verify counts and membership, not the correctness of source interpretation. Selected second source readings provide bounded computational cross-checks; source locators make the interpretations inspectable. No independent human-coded agreement estimate or controlled comparison of analysts using this procedure versus existing guidance was collected. Retrospective applications with visible published outcomes cannot estimate decision accuracy or incremental analyst benefit.

The supplement's *Result-record index* and *Current-code reconstruction* retain every record, locator and current code. Publication status and the inspected version remain distinct: a preprint contributes source-scoped evidence, and an artifact's advertised result is not a measured result. Bibliographic syntax or identifier shape does not authenticate a source; unresolved identity, access or version equivalence remains pending in the saved records.

3. What a Result Can Support

In the original empirical ledger, one result record describes an evaluated target, receiver, intervention and outcome package. It is a comparison unit, not an independent trial. Changes of primitive, execution mode, target/receiver contract or separately measured cost object require distinct records; model, workload and parameter choices remain coordinates within a record. Numerical joins preserve those coordinates. A source-described common configuration can link separate evaluations without establishing identical requests or paired runs. The card retains the strength of that linkage, as well as observed mismatches between measurement objects.

A comparison is compatible for the requested object when required fields align, incompatible when an observed object conflicts, and unresolved when the necessary binding is missing. A token-recovery shortlist does not require equal costs; a cost comparison does. The source-level record identity, rather than a family name, determines which comparison is permitted.

TABLE 2. A SINGLE RESULT RECORD. ONLY FIELDS RELEVANT TO THE REQUESTED COMPARISON MUST ALIGN; UNAVAILABLE REQUIRED FIELDS REMAIN UNRESOLVED.

Record field	Required meaning
Source	Family/result ID, version, source locator, reading scope and hash.
Target	Hidden property, inference goal and protection relation: confidentiality, attribution, human/agent distinction or authorized diagnosis.
Receiver	Actual permission and released representation, including session/document linkage and any auxiliary inputs.
Intervention	Where protection ends; whether edits affect stored traces, transmission or executed actions.
Attacker	Fit/calibration condition, test population, training access and adaptation budget.
Workload	Task, system, topology/hardware and grouping/holdout.
Outcomes	Inference metric / denominator; objective or proxy utility; applicable cost object; uncertainty and field states.

TABLE 3. COMPARISON RULES. A KNOWN MISMATCH AND A MISSING BINDING HAVE DIFFERENT CONSEQUENCES.

Requirement	Condition and resulting license
Descriptive comparison	Identify the shared target and receiving view. Describe differing populations or metrics separately; numerical cost comparisons additionally require aligned cost objects. Token payload and wire bytes cannot be ranked as the same cost.
Paired effect	Compare the same defended test units and outcome. A known test-unit mismatch makes that subtraction incompatible; an unverified fit binding leaves the effect unresolved.
Deployment transfer	Establish a bridge for the changed receiver, intervention, workload, topology or hardware. A demonstrated violation blocks the proposed transfer; an untested bridge leaves it unresolved. Permit only the range and endpoints actually covered.

The 39-record ledger contains 31 privacy/attribution mitigation records, two AHB evasion records and six contextual controls. The historical 20-family/33-record mitiga-

TABLE 4. TAXONOMY OF CHANGES RELEVANT TO EVIDENCE REUSE.

Changed object	Typical transition	Evidence to carry or renew
Observed signal	Receiver, released representation, hidden target	Keep the target and permissions; measure the actual receiving trace.
Executed intervention	Prefix/semantic cache; transformed trace/live scheduler; read/write cover	Keep mechanism and state boundary; bind function and cost to the executed configuration.
Evaluation setting	Attacker fit, workload, model, hardware or network	Keep training/test provenance; establish the proposed holdout and relevant baseline.

tion inventory includes AHB; the 19-family/31-record privacy/attribution inventory excludes that one family and its two records. The six controls are outside both mitigation inventories. These are nested accounting frames, not competing totals. The six-source defense-first challenge remains separate because its receivers and sampling frame differ. The broader frozen R47 map uses version-traceable papers, with family and configuration relationships recorded separately. Formal statements, designs and empirical results remain distinct evidence objects; pending identities and eligibility are not folded into the historical ledger. Record splits encode the distinctions the audit asks readers to preserve, so split counts and different-object pair counts partly follow from the construction. They are bookkeeping for inspecting individual joins, not independent empirical findings about the literature or a validation of the framework's effectiveness. The supplemental audit trail retains the complete record indices, granularity checks and query reconstruction.

4. A Taxonomy of Evidence-Reuse Boundaries

The taxonomy describes the change between a source result and the proposed deployment. Several boundaries can change together.

Receiving view. The generating layer does not determine the receiver's observation. A network observer, tool provider and private-state-aware formatter can expose different projections of the same execution. An output filter's semantic authorization therefore does not by itself bound timing, length or representation choice. A source-scoped guarantee or attack score is retained only for its named released view.

Affected state. A saved-trace transformation, a live framing wrapper and an admitted tool action are different interventions. Lossless framing can preserve a content projection without another content-quality benchmark. A concurrent read still uses resources; an apparently same-value write can consume stateful permissions or create accounting effects. Task completion, quotas and execution cost require their own operating point.

Attacker and workload. Protection under a fixed classifier does not identify protection after defended-data fitting. Likewise, evaluating a rule on several models differs from transporting a source calibration unchanged. A workload change can affect attack training, defense parameters, quality and resource costs separately. The evidence card records what remains fixed and which object is measured again.

Relation to established evaluation methods. Website fingerprinting already supplies adaptive attacks, cost trade-offs and environment-shift tests [9], [5], [6]. Robustness guidance and benchmarks standardize attacker access and evaluated interfaces [7], [8], [10]. Adaptive testing, workload separation, observer specification and protection/cost trade-offs are inherited principles, not conceptual inventions of this SoK. The application here makes LLM/agent-specific joins inspectable: prefix-policy protection versus semantic-cache processing cost, authorized message content versus an exposed score or form, and a discarded reply versus the state of an executed tool call. The external WF examples reproduce decisions already supported by their source papers. The added material is a version/locator-linked organization of these objects and bounded experiments, not evidence that the procedure finds errors missed by prior guidance or improves analyst performance. Formal DP, information-flow, constant-time and oblivious-memory results retain their stated adjacency, leakage model and runtime assumptions [11], [12], [13], [14]; an empirical classifier score is not their substitute.

5. Configuration Decisions from Defense Evidence

5.1. The question is which configuration the evidence supports

The C1–C5 attack-evaluation frame links 19 privacy/attribution families to 31 mitigation records, including attack-oriented and defense-oriented studies. Two AHB evasion records and six contextual controls remain outside this synthesis. These selected records support inspection of particular evidence packages; their reporting counts do not estimate how often defense research measures utility, cost or adaptive resistance.

R8 resolves the claimed function before asking for a quality score. Changed prompts, sampling, approximate computation or component coverage need matched quality evidence. Lossless framing, fixed-trace transformations and exact logical preservation instead support a narrow content/function claim. Delay, resources, timeouts and task completion remain separate outcomes for both classes.

5.2. Cache reuse: what can support a candidate configuration?

CachePrune excludes key–value (KV) cache entries marked sensitive from cross-user sharing. Its v1 Table 1 reports QASPER direct exact recovery of 0% and contextual exact recovery of 3.21%. Table 2 reports default QASPER F1 of 0.28 versus 0.30 without sharing, and time to first token of 273 versus 797 ms, using Mistral-7B on a single

TABLE 5. CLAIM-CONDITIONED REUSE RULES. EACH ROW NAMES THE REQUESTED INFERENCE AND THE EVIDENCE NEEDED TO SUPPORT IT.

Item	Reuse query → required binding (**B**)
R1 Target	Same protected interest → explicit hidden property, inference goal and authorization relation.
R2 Receiver	Same observer protection → permissions, released view and auxiliary inputs.
R3 Intervention	Same executed mechanism → placement/mode: stored data, wire emission or actions.
R4 Protected fit	Resistance to protected-data fitting → identified protected training/calibration and testing.
R5 Matched adaptation	Effect of adaptation → clean/protected fits tested on the same defended population.
R6 Workload	Changed workload/system → tasks, configuration and the relevant holdout axis.
R7 Inference	Quantified protection → metric, denominator and scoped defended evaluation condition.
R8 Quality	Changed generation/computation → matched quality; a preservation witness exempts only that scoped quality query.
R9 Cost	Resource-budget feasibility → units, baseline and cost object at that operating point.

B: bound; **P**: partial binding; U: unreported in inspected scope; A: source unavailable; ?: binding unverified; –: inapplicable. For R4/R5, – means that a training-fit contrast is inapplicable; it does not establish that non-learning adaptive attacks are irrelevant or absent. For task timeliness or completion, content preservation alone is insufficient. First determine applicability, then assign evidence status. Family **B** means at least one scoped record satisfies the item, not that every result does.

H100 [4, 3.1,5.1–5.2,5.6]. The direct result is conditional on detector-marked spans and an abstract binary cache-reuse oracle, not measured WAN timing. Detector errors and contextual inference from other tokens remain possible. These source-reported values are descriptive. The default configuration supplies a useful association, but identical requests or paired runs across the protection and serving tables are not established. Its two recovery targets reuse this evidence rather than providing independent validation. The fixed-chunk comparison in Section 5.7 disables privacy filtering and cannot price the protected configuration. Retain the labeling assumptions and operating point when assessing feasibility; serving performance does not price every deployment resource.

Early Bird's minimum-shared-prefix rule instead uses measured hit/miss classifiers to parameterize a simulated oracle and a K-token predictor. Its v5 Table VI reports token recovery of 91.5% at $K = 1$ and 49.0% at $K = 4$ [1, V-A]. The approximately 6 ms overhead in §V-B belongs to a separate semantic-cache deidentification prototype, evaluated on 1,000 Presidio sentences. It cannot price the prefix rule. We did not locate deployed-serving cost or matched task utility for that rule in the inspected mitigation section. Both mechanisms can enter a threat-specific shortlist, but their recovery numbers are not a controlled comparison. The prefix candidate needs its own serving-cost and functional measurements.

Other source bindings preserve different functional objects: SparLeak links reconstruction and ROUGE to replacement intensity, while TDXRay measures vocabulary coverage [15], [16]. The supplement's *Current-code reconstruction* retains their source witnesses and partial bindings.

For NetEcho/Whisper framing, SPEC aggregation and OTRO's token preservation, a structural witness supports only the stated content/function invariant [2], [17], [18], [19]. SPEC and Dangol also combine scoped preservation with measured inference and cost, but payload, capture, loopback and serving costs answer different budget questions. None can substitute for an unmeasured deadline or task outcome.

Separate components and unresolved timing evidence. SpliceDefense v1 reports nonzero signed residuals for constant-time boundary fusion (CTBF), including a bulk median of -9 ms; their sign alone establishes neither timing indistinguishability nor a successful adaptive attack [20, VII-C, Table II]. Quantized chunk padding (QCP)'s reported sub-1% prefill cost concerns another component, not a paired CTBF or combined-deployment cost. The local simulated-tenant experiments assume logical co-residency, and the longer extraction cases deliberately include the true token in a 21-candidate set. Table II's 15 tasks/75 steps and the separate ten-token targets remain unreconciled. Defended adaptive distributions, task quality and combined cost remain pending.

5.3. Hardware cases expose another selection boundary

SecEmb was already adjacent in the original map. It replaces embedding-table access with deep hash embedding and argues protection against its stated memory/control-flow observer. Its GPT-2 medium/OpenWebText results report minimum perplexity 15.0 versus 14.6 for lookup and generation overhead of 2–5%, depending on batch size [21, V-B,VI-D]. The inspected PRIME+SCOPE attack demonstration instead uses DLRM; it cannot supply a GPT-2 recovery score. Exact checkpoint/run linkage of GPT-2 quality and latency also remains unverified. This is useful defense and functional evidence that an empirical-attack inclusion rule can leave outside the core sample.

Palermo v1 considers a trusted on-chip system and an external-memory/bus observer, using a simulated GPT-2/OpenORCA workload [22, II-B,VI–VIII]. Figure 9's mutual-information statistic concerns binary stash/tree state and median-split response timing; its near-zero value is not a prompt-recovery result or a general zero-leakage bound. Modeled execution performance and synthesized hardware costs are separate evidence. Output quality was not located in the inspected evaluation sections, and identical runs across the security and performance figures were not established. This case supports an observer-specific hardware tradeoff, subject to those limits. These selected examples neither complete hardware coverage nor estimate omissions; they leave the frozen ledger unchanged.

Retain a conditional guarantee at its stated boundary. OTRO v1 states DRAM-trace indistinguishability for equal numbers of tokenizer table accesses under its ORAM and pseudorandom-function assumptions; computation timing

and other implementation effects are excluded [19, III,V–VII, Appendix B]. Access count, or padded epoch count, and other public quantities remain visible. Its uniform 20,000-prompt closed-world leakage exercise is not a universal bound. Plotted TTFT excludes access-count padding, and the cost text contains unresolved discrepancies. These are separate objects, not a paired, padding-inclusive multi-client protection/quality/cost result. The frozen supplement also retains Opal's stated enclave/ORAM guarantee and an eFPGA LLM countermeasure design as source-version D judgments, with family identity and independent review pending. Their evidence types do not require a scored attack and do not imply end-to-end measured confidentiality.

5.4. Agent-associated interventions need distinct task claims

UI preserves saved action order; Dangol preserves fixed message bodies; WRA returns only the real output despite concurrent research decoys. These support narrow preservation claims. The cloud prompt/temperature interventions can change generation quality. None of these four map-selected families binds an explicit same-condition end-to-end task score, but they do not form a homogeneous sample of agent defenses. WRA is the sole added-call example in this subset; its usefulness judgment uses another condition. The actionable question is whether the particular added execution changes task completion, not whether an aggregate count establishes a field-wide utility gap [23], [3], [24], [25].

5.5. What the selected defense-first sources add

The challenge cards supply positive functional measurements and CacheSolidarity's scoped exact-prefix preservation. They differ from the original sample in selection, receivers and outcomes, so their quality counts are not a comparison of reporting rates. A functional measurement also need not form a joint protection/function/cost package (Table 6).

OCELOT links protection, task success and mediation cost at its semantic release boundary; its ledger invariant and calibrated inference guarantee remain distinct. AirGapAgent binds disclosure and answer utility for third-party contextual extraction. SICC explicitly isolates form-only inference, and DP-MemView separates its compliant semantic interface from an internal-score diagnostic. These observations require receiver-specific judgments. PlanTwin's unrestricted-field diagnostic cannot be joined to budgeted planning quality. The cards therefore guide different decisions even where functional measurements are available.

R4/R5 concern protected-data attacker fitting and the clean-versus-protected fitting contrast on the same defended population. They do not inventory all adaptive evaluation. SICC and OCELOT test broader defense-aware attacks without satisfying that narrow R5 query. An unreported or unverified fit binding cannot establish that an attack was not evaluated. The supplemental inventories and sensitivities preserve all counts as descriptions of the coded sources; neither small denominators nor record splits establish the prevalence of a deficiency or the effectiveness of this framework.

SoK Takeaway 2 (Bind function and cost to protection). Resolve the requested function first. Retain a source-bound preservation witness for its stated object; use matched quality evidence for changed content or behavior, and a task outcome for added execution. Reuse protection, function and cost together only where their receiver and evaluated configuration align.

6. Which Evidence Supports Defense Transfer?

6.1. Separate the evaluated object before counting holdouts

The transfer audit distinguishes three questions: does an *attack* recognize new inputs or systems; is a *defense condition* evaluated; and does a *fixed attacker or defense calibration* carry from a specified source to a target? Table 7 makes the distinction operational. Repeating a defense on several models can establish condition-specific outcomes without establishing a held-out model or transferring a fitted calibration.

The 82-family discovery matrix covers task/input/user, model, harness/topology, hardware, network/load and time. H means reported held-out separation; V means condition variation/recalibration, X a scoped restriction, and ? unresolved extraction. Of those 82 families, 13 have task/input H, three model H and two topology H; the axes can overlap and their counts must not be added. Four of the 13 task/input families belong to the historical 20-family mitigation inventory. The three model-H families lie outside that inventory; the two topology-H families are privacy/attribution mitigations whose held-out objects are workflow recognizers. These five distinct model/topology families (3+2) provide attack-generalization evidence, not verified source-frozen defense-calibration transfer. This is a statement about those five entries, not all 82 families or all defenses. The 13 task/input entries need individual defended-condition linkage. The supplement retains the family identities, source locations and full six-axis matrix.

6.2. A changed generator and a tested defense are different bindings

LLM-Fin, ArchSnoop and LeakyLMs evaluate recognition of unseen model variants or architectures [32], [33], [34]. This supplies attack-side model generalization. In the traffic benchmark, graph-to-star and star-to-graph classification explicitly move the fitted attacker across topologies; the prompt/temperature countermeasures are separate evaluations [25, V-B–C]. Dangol's leave-composition-out test likewise examines recognition of a novel workflow shape, separately from its protected-capture frontier [24, 9.8–9.12]. The reusable result in both topology entries is the recognizer's tested generalization.

TABLE 6. DEFENSE-FIRST CHALLENGE SET: FOUR REQUESTED BOUNDARY SOURCES AND TWO ADDITIONAL SOURCES FROM A BOUNDED SEARCH. THE SIX SELECTED FAMILIES ARE NOT A DEFENSE CENSUS AND ARE NOT POOLED INTO THE ORIGINAL 19-FAMILY/31-RECORD SOURCE FRAME. EACH ROW RETAINS ITS INSPECTED SOURCE AND RECEIVER.

Source	Receiver / protection evidence	Function retained	Cost / transfer boundary
CacheSolidarity v1 [26]	Cross-user TTFT; 20-probe timing demonstration, not a scored recovery rate (§5.2.3, Fig. 9).	Exact-prefix logical preservation (§3.2).	TTFT/hit-rate and component costs; model/load variation (§5.2), not frozen calibration transfer.
PlanTwin v2 [27]	Cloud-visible schema; unrestricted-field re-identification, not budgeted protection (§5.6).	Planning-capability quality (§5.2, Table 5).	Heuristic-default versus SLM timing requires a configuration match (§5.4, Table 6).
SICC v1 [28]	Isolated length/structure-form inference and adaptive transcript attacks (Tables 3, 5–6).	Protocol/model receiver completion (Tables 7–9).	No compiler LLM calls; measured runtime cost remains partial. Authorized-view scope is retained.
DP-MemView v1 [29]	Semantic transcript inference; separate exposed-score diagnostic (AUC 0.976; RQ2).	Matched response quality/personalization (Table 2).	Runtime cost unbound; corpus-text transfer fixes scaffold/policy and reuses the auditor (C.3/C.5).
OCELOT v1 [30]	Semantic trajectory inference; held-out stress proxies and adaptive attacks (§V).	Condition-matched task success (Table II).	18.3% wall-clock overhead (§V-F); ledger arithmetic and calibration-relative privacy are distinct.
AirGapAgent v2 [31]	Contextual extraction/disclosure, not passive metadata inference (§5.3; Tables 3–6).	Appropriate-answer utility (Tables 3–5).	Runtime cost unbound; model/scenario variation does not establish frozen calibration transfer.

All five quality-applicable cards contain a functional score; the remaining card supports narrow preservation. These selected-source observations do not estimate field prevalence. Within these six cards, strict R5 has four inapplicable and two unreported entries; this is not a global count of adaptive evaluations. SICC and OCELOT explicitly evaluate broader adaptive attacks. DP-MemView's exposed-score diagnostic is not its compliant response view. CacheSolidarity becomes PrefixWall in v2; that title relationship does not merge version-specific results. SICC date metadata and exact publisher-version equivalence remain unresolved where recorded. Artifact README expectations are not paper measurements.

TABLE 7. THREE EVIDENCE OBJECTS FOR INTERPRETING HOLDOUTS. THE SAME FAMILY MAY SUPPLY SEVERAL OBJECTS IN SEPARATE EXPERIMENTS.

Object	Required link	Permitted reuse
Attack generalization	Training/profiling source → held-out inputs or systems; attack outcome.	An attacker or recognition method covers the stated target domain.
Defended condition	Named mechanism, local fitting, defended test and outcome.	Retain that operating point; multi-model testing alone is not held-out transfer.
Defense transfer	Source-selected policy/calibration held fixed → target; local-calibration comparator and attacker fit specified.	Transport the evaluated policy only over the tested direction and scope.

There is nevertheless useful *defense-side variation* evidence. Time Will Tell v1 tests a 60-word-explanation instruction on three selected high-bias classification tasks with unchanged three-shot examples and perfect output-token-count observation. Reported mean attack success rate (ASR) changes from 88.2% to 61.9% for GPT-4o and from 79.3% to 80.3% for Gemma2-9b-IT [35, VII-B]. This supports condition-specific variation under a common instruction, not an adapted remote-defense test. Separate padding estimates do not establish matched protected task quality and cost. Whisper Leak evaluates mitigation conditions across provider-models, while its prompt holdout describes the classifier protocol [17, 3.2,6]. Preserve those per-condition results; a shared rule, a refitted local attack, a source-trained attack and a source-calibrated defense are four distinct comparisons.

Deployment decision. For the proposed change, name the source, target, held-out unit, quantities held fixed or refitted, and protected outcome. Carry forward only the corresponding binding. An attack-generalization result justifies including that attacker at the target; a defended-condition result supplies its local outcome; a source-to-target defense-calibration result supports only the stated transfer. This prevents both overclaiming portability and discarding useful local evidence. Unresolved hardware/time entries in the discovery matrix identify source follow-up, not absent experiments.

7. Synthesis: From Evidence to Reuse Decisions

The record atlas and source cards organize evidence requests for a proposed reuse. Three worked cases illustrate receiving, execution and fitting distinctions; they do not measure whether analysts make better decisions using the procedure. The transport and online utility studies extend the source examples with measurements of the specific boundary under discussion. The cache decision in §5.2 illustrates how source evidence supports a candidate while leaving a precise measurement unresolved.

7.1. Case 1: bind the final target-bearing emitter

NetEcho's upstream packet-size shaping operates on a different projection from the tool-call sequence. Its plaintext-aware frontend then emits a new size trace

TABLE 8. PROPOSED CLAIM-CONDITIONED REPORTING DECISIONS DERIVED FROM THE RECORD AND HOLDOUT AUDITS. THE TABLE SPECIFIES EVIDENCE REQUESTS; ITS ACCURACY, ANALYST AGREEMENT AND ADVANTAGE OVER EXISTING GUIDANCE HAVE NOT BEEN EVALUATED.

Reuse question	Evidence decision	Retain or measure next
Emitter / representation	An upstream or buffered capture can differ from the final receiving trace and delivery schedule.	Bind inference to the final receiver; retain the emitter's overflow, byte and delivery-delay scope (R2/R3/R9; §7).
Read cover → writes	Discarding a reply does not establish unchanged commits, authorization or workflow outcomes.	Bind the protected state and utility to the executed intervention; include cover calls, quotas and failures (R3/R8/R9).
Customer / model calibration	A reported split can separate inputs without changing the generator or calibration population.	State held-out units and calibration provenance; separate local fitting variability from source/domain changes; evaluate the requested transfer axis (R5/R6; §6).
Attacker observation	An outcome for one projection or fitted attacker does not supply a result for a richer receiving view.	Evaluate comparable defended tests under the intended observation and adaptation contract; retain receiver-specific inference metrics (R2/R5/R7).
Stored trace → online path	A trace transformation or simulated span does not measure executed queues, completion or online latency.	Bind the receiver, harness and resource denominator to the online condition; establish the relevant topology/network holdout (R3/R6/R9).
Cache-reuse candidates	A source-described configuration can link evidence without paired runs; a cost from another cache primitive is incompatible.	Retain the documented configuration and linkage limits; measure prefix-policy serving cost separately, and re-evaluate changed workload/hardware (§5.2).
Component → workflow utility	Component quality or logical state conformance does not determine a planner's response to scheduling and deadlines.	Report utility at the requested workflow boundary and the same intervention/configuration as protection and cost (R3/R8/R9).

(grouped token counts or character lengths), reopening a size channel after upstream protection [2]. SplitDecoy is a separate map-only return-path example, outside the 39-record mitigation ledger: the cloud sees nonzero gradient row support on every tested frame across nine seeds [36, Sections 5.1–5.3]. Both change the receiver boundary and therefore require a separately bound inference result.

SICC further isolates length/structure-based inference after message content is authorized [28, Table 3]. DP-MemView's separate diagnostic exposes an internal score while holding compliant view-selection and response paths fixed, yielding an attribute AUC of 0.976 [29, RQ2, following Table 4]. This is a different released view, not an attack on the compliant interface. The evidence card therefore retains the semantic privacy result and separately rejects applying it to the exposed score.

7.2. Case 2: bind utility to the action that executes

WRA discards decoy reads for Type I confidentiality; AHB's Type III evasion executes GUI events, a different objective and state contract [3], [37]. A preserved response and an unchanged database therefore establish different functional claims. Our fresh retail-sandbox study tests this distinction with separately executed planner/user interactions and cloned initial states; these share tasks and services and are not statistically independent trials. After the first eligible order lookup, the three arms add nothing, one hidden read, or one hidden same-item write. The write's response is discarded, but its payment-ledger and modification-quota effects remain visible to subsequent execution.

In the mutable Qwen service-alias cohort, 187 three-arm groups cover 98 tasks selected nonrandomly from incomplete GLM blocks. The selected subset retains the protocol's randomized execution order and arm permutations; this does not make the cohort representative. The 113 complete GLM groups remain a separate exploratory description. Averaging each task's available repeats before giving tasks equal weight, write-cover success is 46.9 percentage points below baseline (95% task-bootstrap interval $[-56.8, -36.9]$). The read-cover difference is $+0.7$ points with interval $[-4.9, 6.1]$, leaving small gains or losses unresolved. This is a deliberately state-changing utility stress condition, not a measurement of WRA's privacy protection. All 97 executed cover writes consume the sandbox's modification eligibility; payment-ledger and quota semantics make the direction of harm unsurprising. The study does not isolate how much of the difference each mechanism causes, test an online side-effect-free write control or estimate a general cost of write cover. Earlier reference-program shadow checks do not provide that missing fresh-planner control. Recovery changed concurrency from 10 to 2 and the transport path; ten interrupted request waits and server outcomes remain unknown. These limits accompany the numerical difference. Appendix A retains all three arms, the paired tests and the cohort boundary.

7.3. Case 3: bind adaptation, calibration and cost to the tested setting

The replay study uses three pinned local model artifacts, 100 synthetic topics with five templates each, and a 128-token generation budget. The binary task distinguishes 50 medical from 50 nonmedical topics, with five topic-held-out folds and a 50% chance balanced accuracy (BA). Saved output chunks are emitted through timed TLS loopback. This finite setting does not establish outcomes for cloud services, real-user topics, WANs or defenses operating during generation.

For each defended test view, D_0 fits on clean training data and D_2 on defended training data. At Qwen 8B's default four-chunk batching point, the matched three-learner raw reference is 79.0% BA, versus 50.0% for D_0 and 73.2% for D_2. Each condition separately selects its best logistic regression, random forest or gradient boosting result on the same five-fold out-of-fold predictions used for reporting. These are exploratory condition-wise suite maxima, not a paired effect estimate for one fixed classifier. Selection can make them optimistic; neither nested selection nor a separate evaluation set is used. Moreover, D_0 logistic regression has AUROC 0.678 despite BA 50.0%: the thresholded result does not imply absence of discriminative information.

The calibration audit also limits the claim. Default bucket, batch and fixed-cell parameters are identical across all source folds, so their zero imported-minus-local differences are degenerate comparisons with no evidence about transferring a changed calibration. Some hard-cap source/fold comparisons have different parameters, while others remain identical. At the reported precision, five of the six conditional intervals include or touch zero; only Llama 1B to Qwen 8B excludes zero, at $+3.4$ attack-BA points with interval $[0.8, 5.4]$. Higher attack BA means worse protection for that imported cap under this comparison, not improved defense utility. The secondary Holm adjustment covers two imports per target, not all targets, policies and points. Appendix A reports all six and the omitted fitting and selection uncertainty. The evidence neither establishes a robust transfer effect nor demonstrates equivalence or absence of a source effect. No reported local fitting-variability control separates a source change from finite-sample calibration variation.

Cost must follow the same configuration. For those 500 Qwen 8B batching replays, response application TLS bytes total 470,239 versus 1,234,823 for paired raw captures ($0.381\times$), because aggregation reduces framing and record overhead. Paired completion differences have median $+0.152$ ms and 95th percentile $+2.381$ ms; the 95th-percentile delay of original chunk endpoints is $+65.730$ ms. These are local replay measurements over 100 topics, excluding requests and handshakes. They do not price online generation or a WAN. The next deployment measurement is an adapted attacker and a nondegenerate, locally controlled calibration comparison, with inference and cost bound to that same execution. The appendix and supplement retain the incomplete original run and separate recovery-arm provenance.

7.4. Bounded reappraisal of published results

The existing external ledger covers 15 measurement records for 14 defense–study pairs in Deep Fingerprinting, WFDefProxy and Mathews et al.; these were retrospective applications with the published outcomes visible, not new experiments or independent predictions. Two examples make the assessment inspectable. WFDefProxy's FRONT comparison reports DF TPR 72.06% versus 42.79%, FPR 2.16% versus 2.49%, and data overhead 52% versus 75% [38, 6.2, Table 10]. The retained assessment keeps each observed protection/cost tuple; it does not attach simulated cost to a live outcome or rank protection at a common FPR. Mathews et al.'s DynaFlow live result reports 379% mean per-site time overhead, but changed URLs and crawler prevent treating its simulation/live difference as a matched causal effect [6, IV-F, Table II; Appendix B].

These distinctions were already supported by the cited papers. The LLM application makes the analogous join explicit: Early Bird's approximately 6 ms semantic-cache de-identification cost does not price its prefix rule, while CachePrune's default-configuration association still does not establish paired protection/serving trials (§5.2). The concrete output is a source-located acceptance or rejection of that particular join, not discovery of a new source-paper error or evidence that analysts using the card outperform existing practice. The full external ledger, including favorable scoped results, remains in the supplement.

What the three findings add. The contribution is an auditable organization of source-specific comparisons: which observation a result covers, which state a tool action changes, and which fitting or calibration quantities differ. The source examples show what a reporting card records; E4 and R44 supply bounded measurements of selected cases. They do not establish new general evaluation principles, reliable analyst judgments or improved deployment outcomes. Those claims remain untested.

7.5. A deployment query determines the evidence to retain

Start with the intended receiver, protected outcome and intervention (Figure 2). Record-level comparison identifies whether the cited result measures that object. Configuration-level joint binding identifies which inference, utility and cost outcomes share its configuration. Deployment decisions use the holdout matrix to identify evidence for the proposed change and specify the remaining measurement. Table 8 proposes the following evidence requests: a new emitter needs a receiver-specific capture, writes need state/authorization checks, source-calibration comparisons need a local fitting-variability control, and an online scheduler needs its own planner utility and latency measurement. The calibration control follows from the comparison design: changing the source can also change the finite training sample used to fit the defense.

Appendix A retains the verified transport and online utility projections. The supplement retains the earlier controlled studies, filled reporting cards and external WF applications.

8. Implications and Ethics

8.1. Deployment and reporting implications

Bind each result to its receiving view. Access-network operators, VPC monitors and receiving tools expose different observations; remote paths can reveal destinations

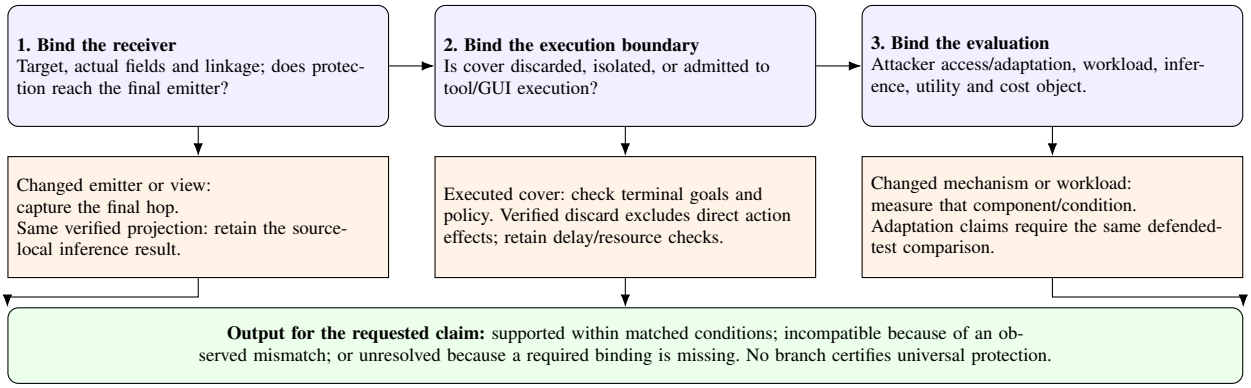


Figure 2. Decision path for a proposed defense reuse. The output identifies evidence to retain, an observed mismatch or a specific unresolved measurement.

TABLE 9. REUSABLE REPORTING CARD. FILL EACH EVIDENCE CELL WITH A VERSION AND LOCATOR OR "UNKNOWN"; THE FINAL ACTION IS CONDITIONAL ON THE PROPOSED DEPLOYMENT.

Field	What to enter
Receiver/target	Hidden property, accessible signal and permissions.
Intervention	Primitive, execution mode, state and trust boundary.
Attacker/test	Training or calibration condition; defended test population.
Function	Scoped preservation witness, component quality or task outcome; shared configuration.
Cost	Resource, units, denominator, baseline and included operations.
Extension	Changed workload/system axis; available holdout; retain, separate or measure next.

even with Encrypted ClientHello [39]. Ingress evidence does not automatically cover host instrumentation or downstream logs.

Bind state effects to the executed intervention. Read queries, isolated execution and gateway cover have different contracts. Dry-run, sandboxes and rollback have limits [40], [41], [42]; idempotence alone is insufficient, and sequence changes can survive rollback [43]. Record the trust boundary, authorization, quotas and protected outcome with each result.

A usable reporting card. Table 9 records six fields and a proposed deployment decision. Filled examples and an editable template retain source locations and unknown states.

Unmeasured validity. Analyst benefit remains untested. Assessing it requires a controlled comparison with existing guidance and independently assessed judgments. Stronger deployment claims need fixed models, a justified cohort, additional domains and a side-effect-free write control for R44. E4 needs independent or nested attacker selection and a local calibration-variability control.

8.2. Ethics and artifact scope

The audit uses public research sources. Evasion and research-intent privacy have different authorization relations [37], [3]; published attack methods may enable inference. The case study uses cloned benchmark databases, simulated users and controlled sockets, without modifying production accounts. Reconstruction materials include source locators, derived labels, protocols, code and aggregate results. Privacy, data-rights and service-term decisions for traces and third-party material remain unresolved; public access does not grant redistribution permission.

Generative AI use. AI tools supported research preparation, implementation, source reconciliation and prose. Authors remain responsible for the audit codes and interpretations. Generative models supplied the studied streams and simulated conversations. Reconstructing analyses requires the specified archived inputs, model/trace identities, code and configuration hashes, environment and saved predictions; replay is not new task-utility evidence. Fresh execution also needs the stated weights or qualified mutable alias, service access and cloned sandbox state. Protocols retain hardware, budgets and recovery records. Missing access or redistribution rights limits reproducibility.

9. Conclusion

This SoK applies established evaluation principles to source-version, observation, execution and configuration links in LLM/agent side-channel research. Its reporting cards and bounded reappraisals make particular evidence joins inspectable; their benefit to analysts and superiority over existing guidance remain untested. The finite synthetic replay illustrates attacker-fitting sensitivity but does not establish a robust calibration-transfer effect. The nonrandom retail-sandbox cohort quantifies a deliberately state-changing write condition, not a general effect of cover traffic. These scoped outputs retain source assumptions and unresolved measurements without claiming complete literature coverage or validated deployment decisions.

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Appendix A.
Worked Boundary Experiments

These supporting cases illustrate the SoK's evidence boundaries. Arithmetic reconstruction does not remove selection, recovery or external-validity limitations. The supplement preserves source matrices, both operating-point grids and cost definitions.

A.1. E4: finite synthetic replay and conditional calibration comparisons

Pinned qwen3:8b, qwen3:0.6b and llama3.2:1b each generate five templates for 100 synthetic topics under a 128-token budget. The binary target is 50 medical versus 50 nonmedical topics. Each of five test folds holds out ten topics of each class with all their templates; chance BA is 50%. Saved native-chunk schedules are sent through timed TLS 1.3 loopback with byte-identical decoded content. Observations cover response application-record lengths and timing, excluding request and handshake traffic. The controls are buckets, chunk batches, variable-stop fixed cells and a hard application-event cap; overflow-retaining minimum-count padding is a separate ablation. None measures a defense executing during model generation, WAN behavior or real-user topic inference.

Calibration uses training folds. D_0/D_2 separately maximize OOF BA over logistic regression, random forest and gradient boosting, fitted on clean/defended data and evaluated on the same defended folds. The raw, D_0 and D_2 values use the same prespecified learner classes but select their winners separately; they do not report a paired effect or interval for one fixed classifier. Choosing and reporting each winner on those same OOF predictions can be optimistic. Full12 additionally includes metadata/profiling attacks and the majority baseline. These are exploratory suite maxima; the topic bootstrap conditions on fitted attackers and omits model refitting and calibration-sampling uncertainty. Winner reselection among saved predictions does not supply an independent evaluation of model selection.

At the default source comparisons, all 45 source–target–fold combinations use a 32-byte bucket quantum, four chunks per batch, or 32-byte cells at 50 ms ticks, respectively. Batching is fixed by design; the source-based bucket and cell quantiles round to the same values. Their zero differences consequently provide no calibration-transfer evidence. The repeated matrices and bucket display remain in the supplement as identical-parameter controls.

Table 10 reports all six hard-cap imported-minus-local contrasts. Some primary source/fold comparisons have distinct caps: Qwen 8B and Qwen 0.6B differ in three folds but both use 87 events in fold IDs 1 and 3; Llama 1B uses 128 in every fold. The paired topic intervals hold fitted

TABLE 10. ALL SIX DEFAULT HARD-CAP IMPORTED-MINUS-LOCAL PRIMARY CONTRASTS. DIFFERENCES AND 95% TOPIC-BOOTSTRAP INTERVALS ARE IN BA PERCENTAGE POINTS, CONDITIONAL ON FITTED ATTACKERS. THE SECONDARY FIXED-WINNER SIGN-FLIP p_{H} ADJUSTS ONLY THE TWO IMPORTS PER TARGET; IT OMITS REFITTING AND WINNER-SELECTION UNCERTAINTY. NO LOCAL FITTING-VARIABILITY CONTROL IS REPORTED.

Imported source	Target	Difference	Conditional 95% interval	p_{H}
Qwen 0.6B	Qwen 8B	-1.2	$[-2.6, 1.2]$	0.2378
Llama 1B	Qwen 8B	$+3.4$	$[0.8, 5.4]$	0.0112
Qwen 8B	Qwen 0.6B	$+0.6$	$[-2.4, 1.8]$	0.8122
Llama 1B	Qwen 0.6B	$+3.8$	$[0.0, 6.0]$	0.1224
Qwen 8B	Llama 1B	$+0.2$	$[-3.0, 4.4]$	1.0000
Qwen 0.6B	Llama 1B	$+1.0$	$[-2.2, 4.8]$	1.0000

TABLE 11. R44 VERIFIED THREE-ARM UTILITY. EACH ARM HAS 187 EPISODES FROM 98 TASKS; THE SELECTED BLOCKS ARE NONRANDOM. TASK-WEIGHTED SUCCESS, PAIRED DIFFERENCE, AND INTERVALS WEIGHT EACH TASK EQUALLY AFTER AVERAGING ITS AVAILABLE REPEATS. THE PAIRED TESTS ASSUME TASK-LEVEL SIGN SYMMETRY; HOLM COVERS THE TWO SHOWN CONTRASTS. THIS IS A SANDBOX UTILITY STRESS TEST.

Condition	Successes /187	Task mean (%) [95%]	Difference (pp) [95%]	Holm p
No cover	154/187	82.7 [75.5, 89.3]	–	–
Read cover	156/187	83.3 [76.5, 89.6]	+0.7 [-4.9, +6.1]	0.859
Write cover	72/187	35.7 [26.7, 45.1]	-46.9 [-56.8, -36.9]	< 0.001

attackers fixed. Secondary sign-flip tests additionally fix observed winners and assume sign symmetry or exchangeability; Holm covers only the two imports for each target, not all targets, controls or operating points. Neither refitting nor winner-selection uncertainty is included in those tests. A local fitting-variability control is not reported, so these comparisons remain exploratory. At the reported precision, five of the six intervals include or touch zero. The one interval clearly excluding zero is conditional and increases attack BA; it does not establish robust portability or a general source effect. Failure to exclude zero in the others does not establish equivalence.

The original base queue retains 34,440 successful captures, one unknown terminal outcome and 559 never-started jobs. Recovery separately collected the latter and a post hoc 500-session whole-arm repeat for the affected Qwen 8B fixed-cell point. The recovery collector used the same local TLS implementation in a new runtime/batch. Among 495 old/new matched sessions, bytes and record counts agree; the median across topic-mean completion differences is -0.461 ms. This descriptive agreement does not establish exchangeability or eliminate batch effects. The original unknown outcome stays unknown. The proxy/direct-path and concurrency changes below belong to R44, not E4.

A.2. R44: fresh online task utility under added execution

The mutable Qwen service alias supplies 561 fresh episodes in 187 three-arm groups from 98 nonrandomly selected tasks; the 113 complete GLM groups remain separate. The protocol records randomized execution order and arm permutations within the selected subset, not random selection of that cohort. Each arm has fresh planner/user interactions and a separate state clone. Extra calls are selected without gold-action information, and unexposed episodes remain included. All 97 executed cover writes have state effects and consume modification quota. This chosen API therefore supplies a strong state-changing stress condition. R44 has no online side-effect-free write arm to separate generic extra-call effects from its payment-ledger and quota semantics; prior reference-program shadow tests are a different experiment. No fixed-weight, randomized-cohort or multiple-domain replication is established.

The table separates episode success counts from task-weighted estimates. Task-bootstrap intervals condition on the selected cohort; sign-flip tests assume task-level sign symmetry, with Holm adjustment over two contrasts. The read interval is not an equivalence bound.

Recovery changed effective concurrency from 10 to 2 and the transport path. Ten interrupted requests have unknown waiting times and server-side outcomes. Recorded time excludes those unknown waits and does not constitute complete total cost.